
Kaan Mika, PhD

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PROFILE

I am a molecular biologist holding a PhD in life sciences with a strong background in insect chemosensory systems. I'm also an accomplished science communicator and an award-winning photographer. Currently, I'm engaged in post-doctoral research at ETH Zurich, where my focus is on studying memory formation in mosquitoes. Furthermore, I serve as the project manager for the "Insects360" initiative. In this role, we aim to educate the public about the significance of insects in our daily lives and showcase intricate 3D models of insects from our extensive entomology collection. Moreover, I am also a columnist for a prominent Turkish national newspaper writing biweekly science articles for a general audience.

EDUCATION

Post-doctoral Researcher, ETH Zurich, Switzerland | 4/2022- Current

Investigating the role of memory-related genes in *Anopheles gambiae* and *Aedes aegypti* mosquitoes.

PhD in Life Sciences, University of Lausanne, Switzerland | 11/2015-10/2020

PhD dissertation: "Identification of regulatory factors involved in the singular expression of members of an olfactory receptor gene cluster"

Research intern, University of Lausanne; Lausanne, Switzerland | 1/2015-11/2015

Research project: Conducting an RNAi-based down-regulation screen in *Drosophila melanogaster*

Master's degree in Molecular biology, Boğaziçi University, Turkey | 8/2012 - 11/2014

M.Sc Dissertation: "The role of Unzipped in axon guidance and midline crossing in the antennal lobe of *Drosophila melanogaster*."

Exchange Student - Roskilde University, Denmark | 9/2010 - 6/2011

Spring Project: "Influence of RctB- ATP binding and RctB titration in RctB-RctB interaction."

Fall Project: "Neuroendocrine regulation of developmental timing and body size in *Drosophila melanogaster*."

Summer Intern, La Sapienza University, Italy | 7/2010 - 8/2010

Basic molecular biology and cloning techniques

Bachelor's Degree - Molecular Biology and Genetics, Istanbul Technical University |
9/2007 - 6/2012

Dissertation: "Effects of Nuclear Factor 1 family on specific genes responsible for hindbrain development."

PUBLICATIONS

Olfactory receptor-dependent receptor repression in *Drosophila*.

Mika K., Cruchet S., Chai P.C., Prieto-Godino L.L., Auer T.O., Pradervand S., Benton R.,
2021/08. Science advances, 7 (32) pp. eabe3745. Peer-reviewed.

<https://www.science.org/doi/10.1126/sciadv.abe3745>

Olfactory receptor gene regulation in insects: multiple mechanisms for singular expression.

Mika K., Benton R., 2021/09/16. Frontiers in Neuroscience, 15 p. 738088. Peer-reviewed. <https://doi.org/10.3389/fnins.2021.738088>

Targeted molecular profiling of rare olfactory sensory neurons identifies fate, wiring, and functional determinants. (& denotes co-second author)

Arguello J.R., Abuin L., Armida J. & **Mika K.** & Chai P.C., Benton R., 2021/03/05. eLife, 10 pp. e63036. Peer-reviewed. <https://elifesciences.org/articles/63036>

AWARDS

"Prix du Professeur Henri-A. Guenin 2022" award for best PhD Thesis of Center for Integrative Genomics, Lausanne.

Winner of scientific photography competition organized by Swiss National Science Foundation, 2020.

TEACHING

Insect ecology - ETH Zurich (2022)

Cellular and Molecular Biology - University of Lausanne (2016-2020)

Embryology - University of Lausanne (2016-2020)

Scientific paper reading in chemoperception - University of Lausanne (2020)

Genetics - Boğaziçi University (2013-2014)

Molecular Biology - Boğaziçi University (2013-2014)

PUBLIC OUTREACH & VOLUNTARY WORK

- Project manager of Insects360 at ETH Zurich.
- Accomplished science communicator on Instagram. My pages: https://www.instagram.com/slice_of_science/ in English and https://www.instagram.com/bir_dilim_bilim/ in Turkish

- Laboratory assistant for public at Mystères de l'UNIL (2015).
- Photographer&Social Media Manager of Pint of Science 2020 - Lausanne
- Volunteer at On The Ground International to help "Caminantes" in Colombia. (2021)
<https://www.onthegroundnow.org/our-mission/>
- Columnist at Cumhuriyet Pazar - Turkey
- Scifilm Hackathon Lausanne 2018 - Méditation movie https://www.youtube.com/watch?v=NZKPLgrhg5o&ab_channel=SciFilmIt

POSTERS

Mika K., Benton R. Transcriptional interference ensures singular expression of clustered olfactory receptors. European Drosophila Research Conference, 5-8 September 2019, Lausanne, Switzerland.

Mika K., Godino L. P., Chai P. C., Benton R. Identification of regulatory factors involved in the singular expression of members of an olfactory receptor gene cluster. 16th European Neurobiology of Drosophila Conference 2-6 September 2016, Platanias, Chania, Crete, Greece.

Mika K., Zülbahar S., Talay M., Hummel T., and Çelik A. The role of Unzipped in axon guidance and midline crossing in the antennal lobe of *Drosophila melanogaster*, European Drosophila Research Conference, 16-19 October 2013, Barcelona, Spain.

LABORATORY SKILLS & EXPERIENCE

- Molecular cloning and gene editing (CRISPR/Cas9, BAC recombineering)
- Fruit fly and mosquito husbandry
- Immunohistochemistry and in situ hybridization assays.
- Setting up behavioral assays
- dsRNA injections
- Advanced knowledge of confocal and fluorescent microscopy
- Experience in cell culture and Protein interaction assays (Western blot, yeast-one-hybrid assay)
- PCR, qRT-PCR

SKILLS

- Literature scan and scientific writing
- Solid knowledge of R (Statistics, data processing and visualization)
- Knowledge of Python and UNIX
- Advance knowledge of Microsoft Word, Excel, Powerpoint, Keynote
- Great image processing skills in Adobe Illustrator, Photoshop, Lightroom and Fiji.
- Sequence editing programs.

MEDIA

Here is some news about me on the internet:

2023: My collaboration with Heidi.news on a science communication project about mosquitoes

Link: <https://www.heidi.news/sciences/pourquoi-on-etudie-la-memoire-des-moustiques-video>

2023: Swiss National Science Foundation interviewed me for having a successful science communication account on Instagram.

Link: <https://www.horizons-mag.ch/2023/06/01/the-internet-ambassadors-for-science/>

2022-Current: My column where I tell scientific stories with the photographs I took was deemed worthy of being published in Cumhuriyet Pazar, a national newspaper in Turkey. You can reach my column from the links below (The texts are in Turkish).

Link: <https://www.cumhuriyet.com.tr/cumhuriyet-pazar/mutfagimizin-gizli-sefleri-tehlikede-1902184>

2023- To read my interview with Life Science Network on my scientific journey

Link: <https://www.lifescience.net/news/4983/interview-with-kaan-a-passionate-of-insects-and-ph/>

2021- 'When Science Speaks' Podcast by Mark Bayer on science communication and photography.

Link: <https://podcasts.apple.com/us/podcast/a-picture-is-worth-a-thousand-words-about-science/id1439186636?i=1000534222250>

2020 - I was featured at "STEM at home" project

Link: https://www.youtube.com/watch?v=nW7asYA3qyk&ab_channel=STEMatHOME

2019 - Lausanne University made an interview with me about science communication

Link: <https://news.unil.ch/display/1564566638098>

HOBBIES

- Insect & landscape photography
- Science communication
- Yoga&Meditation
- Snowboarding
- Website design & Social media managing

LANGUAGES

- Turkish (Native) , English (C2), French (B1), Spanish (A2), German (A2)